Drone Traffic Fingerprinting

What We Can Detect, Exploit, and Package

Sources: Christof 2021 (DJI Wi-Fi RE thesis) · Behind the Wings 2023 (DJI hijacking) · OcuSync DroneID 2026 · Detecting Drones via Encrypted Traffic Analysis 2019 · Drone Profiling 2018

The Paper's Gaps

Gap	Risk
OUI-only vendor ID	MAC randomization bypasses 40-point weight
No traffic pattern analysis	Encrypted Wi-Fi looks like any high-bandwidth client
2.4 GHz only	Misses 5 GHz and OcuSync drones entirely
Single node	No direction finding or range discrimination
No protocol analysis	Cannot differentiate drone model or flight mode

Layer 1 — 802.11 MAC Layer (ESP32 Today)

Visible from promiscuous frame headers. Zero decryption needed.

Signal	What It Tells You
Source MAC	Device identity (can be randomized)
RSSI	Range estimate, variance for stability scoring
Frame type	Management vs data vs control
Packet size	Biometric: small stick commands vs large video frames
Direction (SA vs DA)	Identifies drone vs controller
Beacon SSID	Drone AP name, if not hidden

Layer 2 — Encrypted Traffic Statistics

Survive WPA2 encryption. 90%+ accuracy from ~170 packets (820ms of traffic).

Feature	Drone Signature
Packet size distribution	Bi-modal: 56B controller to 1472B drone
Traffic direction ratio	Drone sends 10-50x more than controller
Video frame cadence	~33ms intervals (29.4 fps H.264 video)
Burst structure	4-6 MTU frames back-to-back, then pause
Flow continuity	No idle gaps over 200ms in active session
Channel utilization	40-80% duty cycle on single channel

Key insight: A smartphone streaming video sends ACKs both ways. A drone link is nearly one-directional. That asymmetry alone is a powerful classifier.

Layer 3 — Decrypted DJI DUML Protocol

Christof 2021: No application-layer encryption on top of WPA2. DUML payload is plaintext post-decrypt.

Element	Extractable Data
Content type byte at +0x06	Classifies every packet: command/video/camera
0x55 delimiter at DUML Hdr	Firm DJI identifier at payload level
Cmd-set + cmd	Identifies operation: fly, camera, battery, OSD
Telemetry (Cmd 0x43/Set 0x03)	Lat, lon, height, velocity, pitch/roll/yaw, flight status
Battery (Cmd 0x03/Set 0x0D)	Cell voltage, estimated remaining power
Session ID	Tracks all packets in one flight session
Packet counter	Sequential ordering, detects packet loss or injection
CRC-16 Kermit	Non-standard CRC — protocol signature
Video stream (type 0x02)	H.264 1280x720 ~30fps, extractable via FFmpeg

Asymmetric Advantages

1. The drone cannot stop transmitting. A drone in flight requires continuous bidirectional comms. It cannot go silent without landing. The target is always emitting.

2. Video streaming is a forcing function. 1080p at 30fps requires ~4 Mbps via MTU-size frames. No consumer IoT device matches this. Phones have bidirectional ACK traffic — drones don't.

3. The protocol is deterministic. DJI DUML has fixed 0x55 delimiters, known CRC-16 Kermit, fixed field positions. Even within encrypted packets, statistical marker searches yield signatures.

4. Hardware is 1000x cheaper. $5 ESP32 captures all Layer 1-2 data. WPA2 handshake capture is a software addition. Commercial C-UAS starts at $2,000.

5. The attacker cannot spoof the profile. Changing packet sizes, timing, and directionality requires firmware modification. Consumer drones don't allow this. Only custom FPV builds could attempt it.

Proposed 5-Layer Detection Model

Rebalanced weights reduce OUI (spoofable) and elevate traffic-pattern features (unspoofable without firmware changes).

Layer	Weight	Hardware
1. MAC OUI + SSID	20 pts	ESP32
2. Traffic Direction Ratio	25 pts	ESP32
3. Packet Size Distribution	25 pts	ESP32

Layer	Weight	Hardware
4. Video Frame Cadence	20 pts	ESP32
5. DUML Protocol Signature	10 pts	ESP32 + key capture

Original paper: OUI(40) + Rate(30) + Var(20) + SSID(10) = 100

Proposed: DirRatio(25) + PktSize(25) + OUI(20) + Timing(20) + Var(5) + SSID(5) = 100

FPV and Homebrew Drones — ESP32 Blind Spots

Above analysis applies to Wi-Fi drones. FPV drones use completely different RF. The ESP32 cannot see them.

FPV Drone RF Profile

Subsystem	Protocol	Frequency	Detectable By
Control	ELRS, Crossfire, CRSF	900 MHz or 2.4 GHz FHSS	HackRF, BladeRF — not ESP32
Video	Analog NTSC/PAL	5.8 GHz	RTL-SDR + downconverter
Video	Digital (Walksnail, HDZero)	5.8 GHz OFDM	SDR + protocol decoder
Telemetry	MAVLink over serial	915/868 MHz SiK	RTL-SDR narrowband FM
R/C	PWM/PPM/SBUS	Sub-GHz proprietary	Not easily detectable

Critical: An FPV drone emits NO 802.11 frames. The 5.8 GHz video link is the only passive detection vector and it requires different hardware.

Homebrew on Wi-Fi — More Detectable

Trait	MAVLink Drone (Wi-Fi)	DJI Drone (Wi-Fi)
Protocol	Open-source, documented	Proprietary, reverse-engineered
Heartbeat	1Hz always-on, msg ID 0	No heartbeat — continuous telemetry
Message IDs	Known: GPS=33, ATT=30, BATT=147	Known: telemetry 0x43/0x03
Encryption	None — plaintext by design	WPA2, no app-layer encryption
Detection ease	Trivial — any sniffer decodes it	Requires key capture or statistics

Key insight: MAVLink's 1Hz heartbeat is the most reliable drone signature. Every MAVLink drone broadcasts system ID, component ID, and flight mode in plaintext every second. Any Wi-Fi sniffer can identify it with 100% certainty from a single packet.

Drones Without Remote ID

Drone Type	Remote ID?	ESP32 Detection	Primary Vector
DJI Mini 3/4 (Wi-Fi)	Yes	Yes	ESP32 + 5.8 GHz Rx
DJI Mavic 3 (OcuSync)	Yes (in OcuSync)	No	HackRF — DroneID in SDR
Older DJI (Phantom 4, Spark)	No (pre-RID)	Yes	ESP32 traffic analysis
FPV homebrew (ELRS + analog)	No	No	5.8 GHz RF detector + SDR
Homebrew + Pi (MAVLink)	Optional	Yes	ESP32 — heartbeat is 100% ID
Custom FCC module	Depends	Maybe	Multi-modal

Hardest problem: FPV drone on ELRS (900 MHz FHSS) with analog 5.8 GHz video. No Wi-Fi, no BLE, no RID. Detection requires 5.8 GHz RF energy (presence only) or SDR with FHSS tracking.

Multi-Modal Detection Strategy

Sensor	Detects	Cost	Gap
ESP32	Wi-Fi drones, MAVLink, Remote ID	$5	Blind to FPV, OcuSync, 5 GHz
HackRF / RTL-SDR (2.4 GHz)	OcuSync DroneID, FHSS control	$30-300	Blind to 5.8 GHz video
5.8 GHz RF detector	FPV video (always-on)	$10-50	Presence only, no classification
ESP32 + SDR combo	Wi-Fi + FHSS + 5.8 GHz energy	$50-100	Full spectrum, single node

Minimum viable stack: one ESP32 + one $10 RTL-SDR with 5.8 GHz antenna + one HackRF. Total BOM: ~$50-150 per node.

Implementation Roadmap

Phase	What We Build	Time
1. Traffic Stats	Packet size histogram + direction ratio. 50% weight.	Week 1
2. Timing Analysis	Inter-packet timing, 30fps cadence. 20% weight.	Week 2
3. WPA2 Key Capture	Log 4-way handshake from promiscuous frames.	Week 2-3
4. DUML Parser	Post-process PCAPs, extract telemetry and video.	Week 3-4
5. Real-time ML	TinyML classifier using all 5 traffic layers.	Month 2

Key References

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4. Bisio et al. (2018). Unauthorized Amateur UAV Detection Based on WiFi Statistical Fingerprint Analysis. IEEE.

5. luyii-code-1 (2026). DJI OcuSync DroneID Research. github.com/luyii-code-1/dji-ocusync-droneid-research.